

Module4Lab2

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1 Module 4 Lab 2: Textbook Problem 3.1

1.0.1 The Bookstore Inventory

Name: Mac Mckay

Course: CS 82A

Repo: <https://github.com/raulmckay-blip/cs82a-portfolio>

1.1 Overview

This lab uses `allbacks.csv`, which has 15 books with their `volume` (cm³), cover `area` (cm², 0 for paperbacks), `weight` (g), and `cover` type (`hb` = hardcover, `pb` = paperback). The goal is to find the median cover type, the mean weight of all the books, and the mean weight for each cover type, then compare the two groups.

1.2 Load the data

```
[8]: import pandas as pd

books = pd.read_csv('allbacks.csv', index_col=0)
print(books.shape)    # expect (15, 4)
books
```

(15, 4)

```
[8]:
```

	volume	area	weight	cover
rownames				
1	885	382	800	hb
2	1016	468	950	hb
3	1125	387	1050	hb
4	239	371	350	hb
5	701	371	750	hb
6	641	367	600	hb
7	1228	396	1075	hb
8	412	0	250	pb
9	953	0	700	pb
10	929	0	650	pb
11	1492	0	975	pb

12	419	0	350	pb
13	1010	0	950	pb
14	595	0	425	pb
15	1034	0	725	pb

```
[9]: books['cover'].value_counts()
```

```
[9]: cover
pb    8
hb    7
Name: count, dtype: int64
```

1.3 Median cover type

Cover type isn't a number, but it still has an order. Paperback is the cheaper, lower binding and hardcover is the higher one. So I set the order as `pb < hb`, sort all the books, and take the middle one, same as a median with numbers.

```
[10]: cover_order = pd.CategoricalDtype(['pb', 'hb'], ordered=True)
covers = books['cover'].astype(cover_order).sort_values().reset_index(drop=True)

n = len(covers)
middle = n // 2          # index 7 = the 8th book
median_cover = covers[middle]

print(list(covers))
print(f'n = {n}, middle position = {middle + 1}, median cover = {median_cover}')
```

```
['pb', 'pb', 'pb', 'pb', 'pb', 'pb', 'pb', 'pb', 'hb', 'hb', 'hb', 'hb', 'hb',
'hb', 'hb']
```

```
n = 15, middle position = 8, median cover = pb
```

Note: The median cover type is paperback.

1.4 Mean weight of all books

```
[11]: mean_weight = books['weight'].mean()
print(f'Mean weight of all books: {mean_weight:.2f} g')
```

```
Mean weight of all books: 706.67 g
```

Note: The average book weighs about 707 g, which is a little over 1.5 lb.

1.5 Mean weight by cover type

```
[12]: group_means = books.groupby('cover')['weight'].mean()
print(group_means.round(2))

diff = group_means['hb'] - group_means['pb']
```

```
print(f'\nHardcovers are {diff:.2f} g heavier on average')
```

```
cover
hb    796.43
pb    628.12
Name: weight, dtype: float64
```

Hardcovers are 168.30 g heavier on average

Note; Hardcovers average about 796 g and paperbacks about 628 g, so hardcovers are around 168 g heavier.

1.6 Checking book size

Before saying the cover is what makes the difference, I wanted to check if the hardcovers are just bigger books. If they were, that could explain the extra weight instead of the cover.

```
[13]: books.groupby('cover')['volume'].mean().round(1)
```

```
[13]: cover
hb    833.6
pb    855.5
Name: volume, dtype: float64
```

1.7 What the difference suggests

The two groups are pretty much the same size:

Paperbacks are actually a little bigger on average (856 vs 834 cm³). Even so, hardcovers still weigh about 168 g more, so it looks like the hard cover itself adds real weight.

1.8 By-hand check

Paperback mean weight:

$250 + 700 + 650 + 975 + 350 + 950 + 425 + 725 = 5025$

$5025 / 8 = 628.125$ g, which matches what pandas got.

Median cover type:

If I line up all 15 books with paperbacks first, it looks like this:

pb, pb, pb, pb, pb, pb, pb, pb, hb, hb, hb, hb, hb, hb, hb

With 15 books the middle one is the 8th, since there are 7 books on each side of it. Counting over, book 8 is the last paperback, so the median cover type is **pb**.

```
[14]: pb_weights = [250, 700, 650, 975, 350, 950, 425, 725]
print('By-hand pb mean  :', sum(pb_weights) / len(pb_weights))
print('pandas pb mean   :', group_means['pb'])
```

```
By-hand pb mean : 628.125  
pandas pb mean  : 628.125
```

1.9 AI Use

I used AI to help with syntax and refine wording on this lab.